Practical No : 10

Aim : Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int request[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;
    int total = 0;
    int visited[8] = {0};

    for(int i = 0; i < n; i++) {
        int min = 1000;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - request[j]);
                if(dist < min) {
                    min = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total += min;
        head = request[index];
    }

    printf("Total Head Movement = %d\n", total);

    return 0;
}
```

Output:

```
Dhanashri@LAPTOP-9O3LMFMK MSYS ~
$ nano pr10.c

Dhanashri@LAPTOP-9O3LMFMK MSYS ~
$ gcc pr10.c -o pr10

Dhanashri@LAPTOP-9O3LMFMK MSYS ~
$ ./pr10
Total Head Movement = 323

Dhanashri@LAPTOP-9O3LMFMK MSYS ~
$
```